Maths — Assignment - 2

1. $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$

$A + B = \begin{bmatrix} 2 & 4 \\ 3 & 3 \end{bmatrix}$ $A - B = \begin{bmatrix} 2 & 2 \\ -1 & 5 \end{bmatrix}$

$A^T = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ $B^T = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$

$(A+B)^T = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$ $A^T + B^T = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$

verified.

2. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$

$3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$ $2B = \begin{bmatrix} 0 & -2 \\ 4 & 0 \end{bmatrix}$

$3A - 2B = \begin{bmatrix} 3 & 8 \\ 5 & 12 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 2 \end{bmatrix}$

$A \cdot B = \begin{bmatrix} 2+0+0 & -1+6+0 \\ 6-0+4 & -3-3+8 \end{bmatrix}$

$A \cdot B = \begin{bmatrix} 2 & 5 \\ 10 & 2 \end{bmatrix}$ dim $\rightarrow 2 \times 2$.

4. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

Hamadard product $\longrightarrow A * B$

$$= \begin{bmatrix} 1.2 & 2.0 & 3.1 \\ 4.1 & 5.3 & 6.2 \end{bmatrix}$$

$$A * B = \begin{bmatrix} 2 & 0 & 3 \\ 4 & 15 & 12 \end{bmatrix}$$

Hamadard Product coincide with standard matrix multiplication in case of when ~~another matrix is a scalar~~ both matrices are diagonal matrices.

5. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times 2}$ $|A| = 4 - 6 = -2$

$3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}_{2\times 2}$ $|3A| = 36 - 54 = -18$

$= 9 \times -2$

$= 9 \times |A|$

$$\boxed{|3A| = 3^2 |A|} \quad n = 2.$$

6. $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 1 & 5 \\ 7 & -2 & 6 \end{bmatrix}$ $|A| = 2(6-10) - 3(24-35) + 1(8-7)$

$= 2(-4) - 3(11) + 1$

$= -8 - 33 + 1$

$= -40 \neq 0$

so A is invertible.

7. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

$A^T = A$

so, A is symmetric.

General $\longrightarrow$ $a_{ij} = a_{ji}$.

8. $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ $\text{Adj } A = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ $|A| = 6-5$
$= 1.$

$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$

$= \frac{1}{1} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$

verifying —

$A \cdot A^{-1} = I$

$\Rightarrow \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

verified

9. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ $|A| = 4-4 = 0$

so A^{-1} does not exist.

geometrically area $= 0$.

10. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $A^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. $|A| = 0-(-1)$
$= 1$

$A^T \cdot A = \begin{bmatrix} 0+(1) & 0 \\ 0 & 0+(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

so $\longrightarrow$ A is orthogonal.

11. $T(x,y) = (2x+y, x-y)$

$T(1,2) = [2.1+2, 1-2] = [4, -1]$

$T(3,-1) = [2.3-1, 3+1] = [5, 4]$

so matrix $= \begin{bmatrix} 4 & -1 \\ 5 & 4 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$

$R_2 \longrightarrow R_2 - 2R_1$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$R_2 \longleftrightarrow R_3$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \longrightarrow R_2 - R_1$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

$R_2 \longrightarrow -R_2$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

$R_1 \rightarrow R_1 - 2R_2$

$A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{Rank} = 2}$

2 non-zero rows.

pivot columns = 1 & 2

13. $(1, 2, 3) = v_1$
$(2, 4, 6) = v_2$
$(1, 1, 1) = v_3$

$v_2 = 2 \times v_1$

so all the vectors are linearly dependent.

as $2 \times v_1 - 1 \times v_2 + 0(v_3) = 0$

also $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{vmatrix} = 1.(4-6) - 2.(2-6) + 3(2-4)$

$= -2 + 8 - 6$

$|A| = 0$.

14. $S = \{(1,2,3), (2,4,6), (3,6,9)\}$ $\quad v_2 = 2v_1$, $v_3 = 3v_1$

so v_2 & v_3 are linearly dependent on v_1

so basis is v_1 i.e $\longrightarrow (1,2,3)$.

therefore it is 1 Dimension.

15. $x + y = 2$
$2x + 3y = 5$ $\Rightarrow$ $\underset{A}{\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}} \underset{X}{\begin{bmatrix} x \\ y \end{bmatrix}} = \underset{B}{\begin{bmatrix} 2 \\ 5 \end{bmatrix}}$

$Adj(A) = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$

$|A| = 3 - 2 = 1$

$A^{-1} = \frac{1}{1} \times \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$

$= \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$

so. $A^{-1} \cdot B = \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

$= \begin{bmatrix} 6 - 5 \\ -4 + 5 \end{bmatrix}$

$X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\boxed{\begin{matrix} x = 1 \\ y = 1 \end{matrix}}$

16. $3x - y + z = 2$

$2x + y + 0 = 1$

$x + 2y - z = 3$

$$\Rightarrow \begin{bmatrix} 3 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} -1 & 2 & 3 \\ 1 & -4 & -7 \\ -1 & -2 & 5 \end{bmatrix}^T = \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & -2 \\ 3 & -7 & 5 \end{bmatrix}$$

$$|A| = 3.(-1) + (-1)(2) + 1(3)$$

$$= -3 - 2 + 3$$

$$= -2$$

$$A^{-1} = -\frac{1}{2} \cdot \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & -2 \\ 3 & -7 & 5 \end{bmatrix} \qquad B = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$A^{-1}.B = -\frac{1}{2} \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & -2 \\ 3 & -7 & 5 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$= -\frac{1}{2} \begin{bmatrix} -2+1-3 \\ 4-4-6 \\ 6-7+15 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -4 \\ -6 \\ 14 \end{bmatrix}$$

$$A^{-1}.B = \begin{bmatrix} 2 \\ -3 \\ -7 \end{bmatrix}$$

Eq (1) — $3(2) + 3 - 7 = 2$ ✓

Eq (2) — $2(2) - 3 = 1$ ✓

Eq (3) — $2 - 6 + 7 = 3$ ✓

$\boxed{x = 2 \quad y = -3 \quad z = -7}$

verified